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SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I and Sem - II
Examination May/June-2026
Engineering Physics

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.no 2, 3, 4, 5, 6 and 7

SECTION – A

Q1 Solve the following questions (any 10)**20**

- A. Electromagnetic waves can travel through:
- a) Solids only
 - b) Liquids only
 - c) Gases only
 - d) Vacuum
- B. What is the correct order of electromagnetic waves in increasing frequency?
- a) Gamma rays, X-rays, ultraviolet, visible, infrared, microwaves, radio
 - b) Radio, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays
 - c) X-rays, gamma rays, ultraviolet, infrared, visible, radio, microwaves
 - d) Microwaves, radio, infrared, visible, gamma rays, X-rays, ultraviolet
- C. In Newton's rings experiment, the central spot is:
- a) Always bright
 - b) Bright in reflected light
 - c) Dark in reflected light
 - d) Colored
- D. The piezoelectric effect is the conversion of:
- a) Mechanical energy to electrical energy only
 - b) Electrical energy ↔ mechanical energy
 - c) Electrical to Mechanical energy only
 - d) Heat to electricity
- E. X-rays can be produced by:
- a) Chemical reactions
 - b) Heating an element to a high temperature
 - c) Photosynthesis
 - d) Sudden deceleration of high-speed electrons
- F. Bragg's Law relates:
- a) Angle of incidence and reflection
 - b) Wavelength of X-rays and interatomic spacing

- c) Intensity and wavelength of X-rays
d) Frequency and energy of X-rays
- G. Electromagnetic waves travel at the speed of light, which is approximately:
a) 3×10^8 m/s
b) 1×10^8 m/s
c) 2×10^8 m/s
d) 5×10^8 m/s
- H. Diffraction and Diffraction Grating, The phenomenon of diffraction occurs when:
a) Waves pass through a narrow opening
b) Waves are absorbed by an object
c) Waves reflect off a surface
d) Waves are transmitted through a medium
- I. The resolving power of a diffraction grating is defined as:
a) The ability to separate different wavelengths
b) The ability to magnify an image
c) The ability to focus light
d) The ability to generate interference
- J. Superconductivity was first discovered in which element
a) Mercury
b) Aluminum
c) Lead
d) Tin
- K. The Meissner effect demonstrates that superconductors are
a) Ferromagnetic
b) Paramagnetic
c) Diamagnetic
d) Non-magnetic
- L. Calculate the longest wavelength that can be analysed by a rock salt crystal of spacing $d = 0.282 \text{ nm}$ in the first order.
- M. The Michelson interferometer splits light into:
a) 3 parts
b) 4 parts
c) 2 coherent beams
d) Incoherent beams

SECTION – B

- Q.2** A) State and explain Michelson interferometer. **5**
B) State and explain magnetostriction effect (generator)? **5**
- Q.3** A) Explain the resolving power of grating? **5**
B) Explain Davisson -Germer experiment **5**
- Q.4** A) State and explain spontaneous and stimulated emission **5**
B) Explain the ruby Laser **5**

- Q.5** A) Explain photo electric effect with diagram? **5**
B) What is superconductivity? Explain the Meissner effect in superconductors? **5**
- Q.6** A) Write short notes. **5**
i. Step Index Fiber: ii. Graded Index Fiber:
B) Explain the application of interference? **5**
- Q.7** A) Explain the theory of Black Body Radiation? **5**
B) Explain time independent Schrodinger equation? **5**